Hide and Seek: Tracing the PRDM9 Locus in Crocodilians Through Synteny

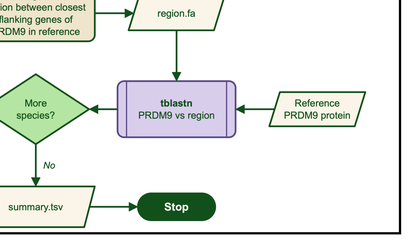
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Background

- What is PRDM9?
  - A gene that dictates meiotic recombination sites across many vertebrate taxons (especially mammals, snakes, and fish)
  - Can give important information on how genetic diversity occurs in a species
- Why crocs?
  - Understudied group; PRDM9 is assumed absent throughout Archosauria (i.e., crocodilians, birds, dinosaurs)
  - Few studies focus on crocodilians despite them being a relatively stable clade (evolutionarily)
- Bioinformatics/ computational approach
  - Combining biology & data science methods



Methods

- Bioinformatic techniques used:
  - Outgroup Analysis with *Chrysemys picta bellii* (the Western Painted Turtle), *known* to have PRDM9, used as reference
  - **Synteny**: comparing gene order between species
- Computational tools used:
  - Python w/shell scripting & CLI
  - BLAST (Basic Local Alignment Search Tool)
  - Tiberius – deep learning model for gene annotation
- Species:
  - American alligator
  - Chinese alligator
  - Australian saltwater crocodile
  - Indian gharial
- Approx. ~10GB of genomic/FASTA data

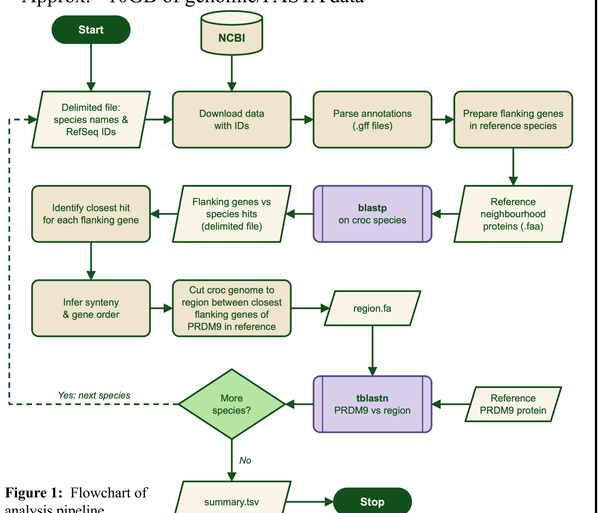


Figure 1: Flowchart of analysis pipeline.

Results

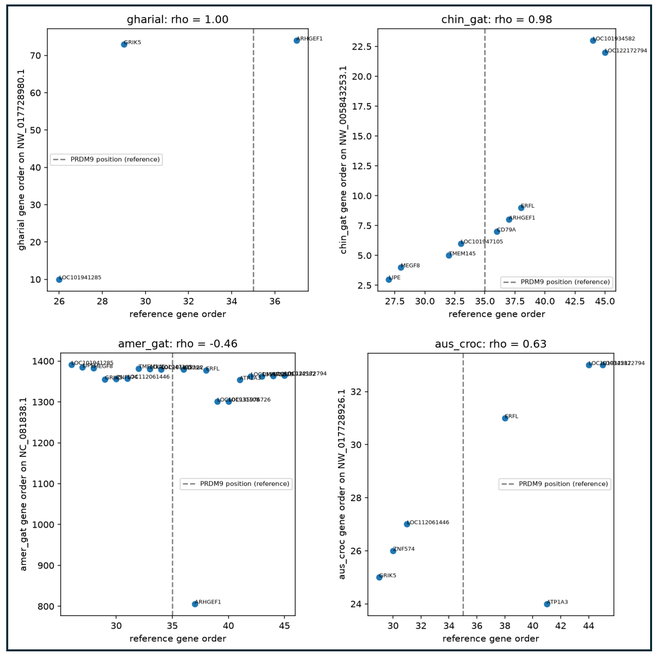


Figure 2: Synteny dot-plots on various crocodilians vs. *C. picta bellii* with labelled identified flanking genes and rho values showing genetic order similarity.

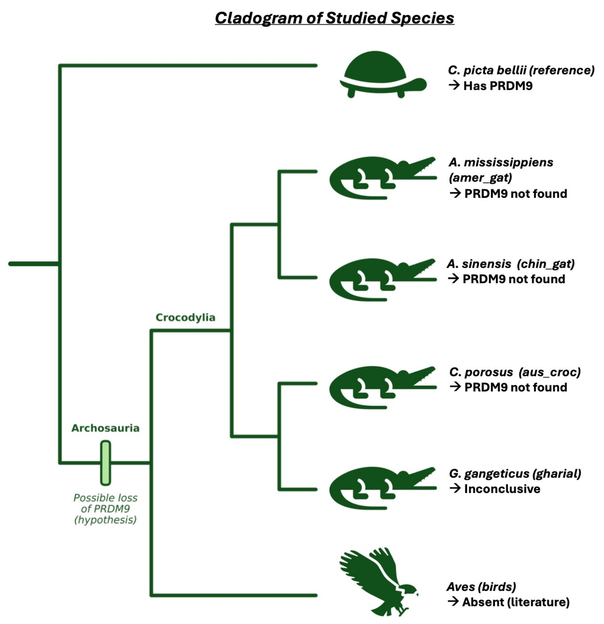


Figure 3: Cladogram depicting relationship between Archosaurs and turtle outgroup, as well as hypothesis for ancestral loss of PRDM9.

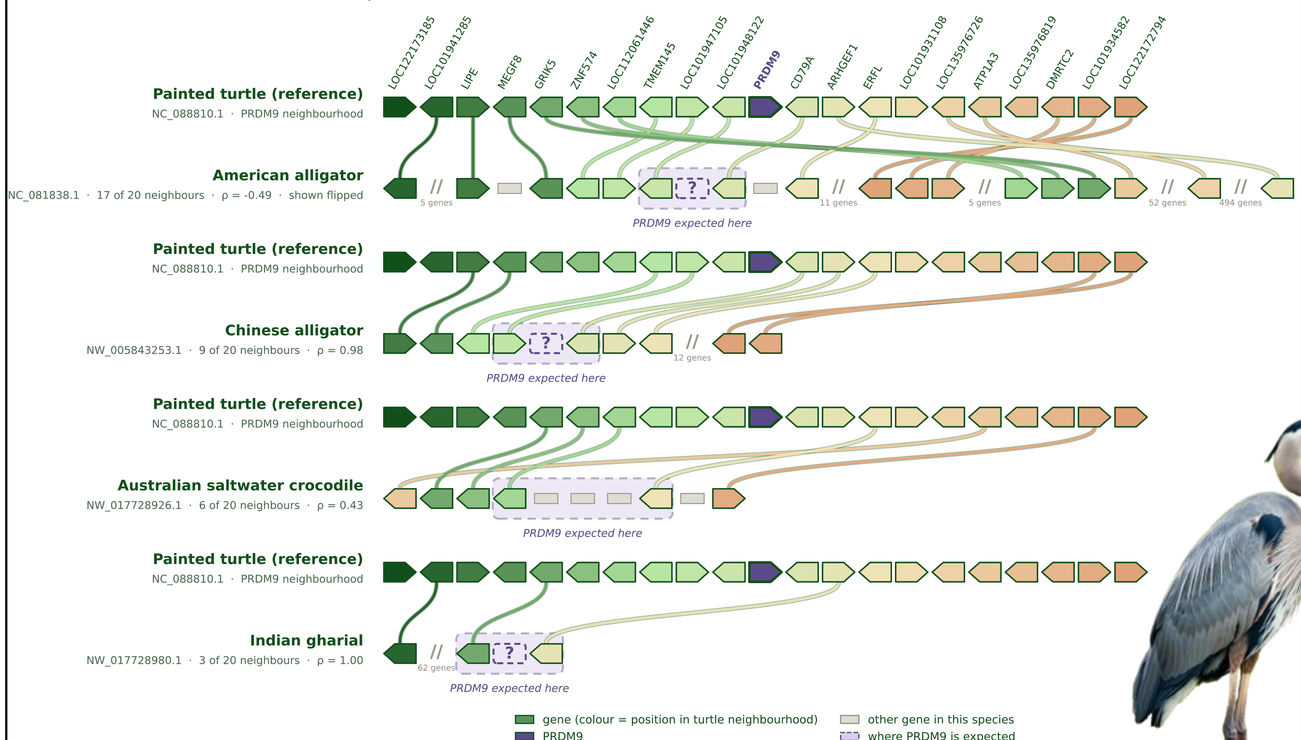


Figure 4: Synteny visualization showing discrete flanking gene matches and how regions of interest were determined. Shows that the American Alligator has the largest number of identified and preserved genes within a scaffold, though this is likely aided by the fact that rAllMis1 is a chromosome-level assembly, rather than scaffold-level.

Discussion

- Among the species investigated, none had confirmed hits for PRDM9 presence
  - Aligns with implications from literature
- If both birds & crocodilians lack PRDM9, it could imply loss stemming from common ancestor

Table 1: Numerical results from pipeline.

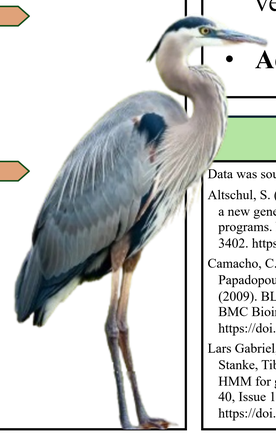
Species	Flanking Genes (of 20)	Synteny (ρ)	Region of Interest Len. (kbp)	PRDM9 Found?
amer_gat	19	-0.46	9.63	No
chin_gat	9	0.98	12.85	No
aus_croc	7	0.63	43.9	No
gharial	3	1.00	398.08	Inconclusive

- **Limitations:**
  - Gharial "results" cannot be corroborated; did not run properly under current pipeline structure (may require manual analysis & oversight)
  - Synteny was determined through non-robust means; weak to paralogs



Future work

- Use more specialized synteny software (e.g., SYNY)
- Emphasize available chromosome-level assemblies
  - Would make results more interpretable & easier to locate
- Consider self-made annotations
  - E.g., Helixer, Tiberius, eggNOG-mapper
- Transfer current and future work to GitHub for version control
- **Add in more genomes/species**



Citations

Data was sourced through NCBI Datasets CLI

Altschul, S. (1997). Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. Nucleic Acids Research, 25(17), 3389–3402. https://doi.org/10.1093/nar/25.17.3389

Camacho, C., Coulouris, G., Avagyan, V., Ma, N., Papadopoulos, J., Bealer, K., & Madden, T. L. (2009). BLAST+: architecture and applications. BMC Bioinformatics, 10(1). https://doi.org/10.1186/1471-2105-10-421

Lars Gabriel, Felix Becker, Katharina J Hoff, Mario Stanke, Tiberius: end-to-end deep learning with an HMM for gene prediction, Bioinformatics, Volume 40, Issue 12, December 2024, btae685, https://doi.org/10.1093/bioinformatics/btae685

GitHub Repository